OPTION THEORY

HAI DANG TRAN NGUYEN
tranngnguyenhaidang21@gmail.com

MAHLÎ, REINETTE
mahli.reinette@etu.univ-lyon1.fr

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M1 Actuariat
ISFA

OPTION THEORY ASSIGNMENT

Part I

Question 2

1 c°

○ Geometric Brownian Motions : $(S^{(1)}, S^{(2)})$

Let : $(S^{(1)}, S^{(2)})$, define by :
— $(\rho, \mu_1, \mu_2, \sigma_1, \sigma_2) \in \mathbb{R}_*^3 \times \mathbb{R}_+^2$
— $(W^{(1)}, W^{(2)})$, 2 brownians, such as $< W^{(1)}, W^{(2)} >= \rho$

$$S_t^{(1)} = S_0^{(1)} exp((\mu_1 - \frac{\sigma_1^2}{2})t + \sigma_1 W_t^{(1)})$$

$$S_t^{(2)} = S_0^{(2)} exp((\mu_2 - \frac{\sigma_2^2}{2})t + \sigma_2 W_t^{(2)})$$

definitions and notations

$$S_t^{(1)} = S_0^{(1)} + \int_0^t \mu_1 S_s^{(1)} ds + \int_0^t \sigma_1 S_s^{(1)} dW_s^{(1)}$$

$$S_t^{(2)} = S_0^{(2)} + \int_0^t \mu_2 S_s^{(2)} ds + \int_0^t \sigma_2 S_s^{(2)} dW_s^{(2)}$$

démonstration

$$dS_t^{(1)} = \mu_1 S_t^{(1)} dt + \sigma_1 S_t^{(1)} dW_t^{(1)}$$

$$\text{Let}: \; f = ln$$

$$dln(S_t^{(1)}) = f^{'}(S_t^{(1)}) dS_t^{(1)} - 12 f^{''}(S_t^{(1)}) d < S^{(1)} >_t$$

2 c°